

**ACCUMULATED CAPITAL WEIGHTED DOLLAR VOLUME AND VOLUME PRICE CONFIRMATION
INDICATOR FACTOR MODEL**

Matt Lutey, Indiana University Northwest, U.S.A.

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ABSTRACT

We outline several volume based technical indicators. The indicators are constructed using deciles of their previous months values and are tradable in real time using data from the Center for Research in Security Prices (CRSP). We test VPCI along with two new volume indicators VPML and VPMCI and show that they outperform a buy and hold on all U.S. stocks with annualized returns of 38.68% and risk of 56.21% compared to the buy and hold return of 13.98% and risk of 66.33% when combining factors from all three models. We also test Accumulated Capital Weighted Volume and Dollar Volume and test a strategy using the 250 day exponential moving average, and 52 week simple moving average. We find annualized return of 10.05% and risk of 12.55%, and 9.58% and risk of 15.80% on both CRSP and the S&P 500 respectively. This is compared with a buy and hold of 9.44% return and 18.77% risk, and 10.18% return and 22.93% risk on their benchmarks. The returns are validated using the asset pricing tests of Kolmogorov Smirnov tests, CAPM and Fama and French 3 factor model and 5 factor model for growth and profitability.

Keywords: Finance, Volume, Wealth Management, Risk Overlay, Capital Weight

1. INTRODUCTION

Volume is shown in academic literature to be predictive of stock prices (Blume et al 1999). Past prices are attributed to predictive ability in stock prices from Lo et al (2000) head and shoulders pattern, Han et al (2014) moving averages. Chang and Osler (1994) and Osler and Chang (1995) also show support for head and shoulders patterns. The patterns are attributed to Edwards and Magee (1948) along with Broadening Top and other nonlinear pattern formations. Pring (2002) provides a guide for momentum patterns, moving averages, volume, among others. Neeley et al (2016) show support for moving averages, volume, and momentum for predicting stock returns. More recently Arvamov et al (2021) use moving average crossovers for predicting returns and find they are more predictive than momentum. Zhou (2016) show theoretical support for using moving averages.

Practitioners who use technical indicators are likely to keep the information to themselves. Practitioners in the field of technical analysis are attributing volume to momentum indicators. Volume Weighted Moving Average (VWMA), Volume Weighted MACD to name a few. Relative Strength Index is a strong technical indicator and smooths out momentum (Hayden 2003). Volatility Bands can be placed around a moving average (Bollinger 1992) and be used for predictive ability. This paper tests the use of popular volume based indicators from Buff Dormeier (VPCI, VPCI RSI, VPML, VPMCI) with the use of multi factor investing (Quantitative Strategies for Seeking Alpha).

We show that the indicators that are derived from volume, volume moving averages, simple moving averages, volume weighted moving averages, relative strength index RSI, and money flow index (MFI) can be predictive and profitable in the cross section of stocks. We create monthly portfolios from the decile values of three main indicators, VPCI, VPML and VPMCI which are constructed from the above noted technical indicators. We select these three indicators and are the first paper to test them rigorously.

The portfolios are formed by the end of the month values for stock prices. The indicators are formed based on their month end decile values which are then invested in the next month. They avoid look ahead bias and are formed using data for the Center in Research in Security Prices (CRSP). The stocks are split, and spin off adjusted.

The indicators that we find work well are VPCI deciles 3-5, combined with VPML decile 6 and VPMCI decile 6. The justification for this is that VPCI is equated to trend, VPML is equated to momentum and from the above noted deciles the middle trend works well. This could be attributed to buying stocks before they reach full strength and selling when they do. Decile 6 of VPML is a risk overlay along with VPMCI.

Additional tests are reported for Capital Weighted Dollar Volume. We create a crossover strategy using the capital weighted volume. We buy when the accumulated capital weighted volume is above its 50 day exponential moving average. We show strong results for this strategy and are the first paper to model this indicator on CRSP. Currently it is available on Optuma for the S&P 500 index only.

We additionally test the VPCI V bottom indicator which is a lower Bollinger Band Breakout of the VPCI at very low values. We combine this with capital weighted dollar volume to create a strong market timing entry signal.

Technical analysis relies on past prices to identify current price behavior and trends that are expected to continue in the future. Literature has been back and forth on the viability of technical analysis. Literature in support of technical analysis include Fama and Blume 1966, Sweeney 1988, Han, Yang and Zhou 2013, Neeley, Rapach, Tu and Zhou 2014, Avramov, Kaplanski, and Subrahmanyam 2018, Osler and Chang 1995, Lo, Mamaysky, and Wang 2000, Blume, Easley, Brock et al (1992), O'Hara 1994.

Campbell (1997) argues the field of finance has seen an increase in the use of quantitative methods. Lo et al (2000) argue that the automation of technical analysis is more difficult to compute algorithmically than these methods which is why they have not (in 2000) been widely adopted.

The paper by Lo et al (2000) uses rigorous mathematical properties to uncover and automate nonlinear visual chart patterns. They argue their methods may have deficiencies in the asymptotic properties of their mathematical algorithms. We address these issues and test whether newer methods namely local polynomial regression and updated methodology for automating the bandwidth selection can uncover the patterns.

Behavioral finance argues that investors have constraints on time, informational, and cognitive ability. This plays a role in the decision-making process. (Simon, 1955).

The efficient market hypothesis doesn't leave any room for technical analysis in current markets. The justification says that markets have already priced in all relevant factors regarding a stock. These include past prices (weak-form), all public information (semi-strong form), and all available information including private and insider information (strong form). The major flaw or drawback is that it relies on the assumption that investors are all rational in their decision-making process and they can obtain all available information prior to making decisions in the markets. Any point in time where this doesn't hold (e.g. a Dentist taking a stock tip from a client, or a college student buying a stock because he wants to gamble his parents money) would lead to some degree of inefficiency in markets at least according to the definitions and assumptions outlined in the efficient market hypothesis by Fama 1970.

There are several papers that outline the theoretical support for technical analysis which relies on markets displaying some form of market efficiency, mainly that they are weak-form inefficient. Treynor and Ferguson 1985, in whether information has been fully incorporated into equity prices, Brown and Jennings 1989 using past prices to gain information in current prices. On the role of volume Blume, Easley, and O'Hara 1994 as well as Grundy and McNichols 1989.

Han, Yang and Zhou 2013 provide justification for profitability in technical analysis studies as being investors not acting on all profitable information. Neeley, Rapach, Tu and Zhou 2014 provide justification for why technical analysis outperforms macro fundamentals in forecasting risk premium in that they pick up on some omitted fundamental variable.

Lo, Mamaysky and Wang 2000, Osler and Chang 1995, Chang and Osler 1994, automated technical analysis. Blume, Easley and O'Hara 1994 study the role of volume. Avramov, Kaplanski, and

Subrahmanyam 2018 study moving averages and their predictability drowning out the effect of a popular momentum strategy. They also provide potential justifications.

Empirical research documenting the inefficiencies of markets include the January effect (Huang and Hirschey, 2006) and the firm size effect (Van Dijk, 2011). Lo and MacKinlay (1988, 1999) on past prices forecasting future returns. Chang and Osler (1994), Osler and Chang (1995).

Lo, Mamaysky and Wang (2000) said that technical indicators can be found using consecutive extrema in stock prices. They also said that consecutive extrema in stock prices are too noisy thus making it impossible to distinguish between informativeness and noise. Their methodology involves smoothing prices to find extrema.

There are a number of factors that contribute to stock returns. Size and value measured by book to market are two that are pointed out in Fama and French 1993. Profitability and growth are mentioned as additional two that could contribute to stock returns.

2. DATA

We use CRSP data monthly 01/01/1990-12/31/2020. Share codes 10 and 11 only for U.S. stocks.

We compute the following volume based indicators: VPCI, VPMI and VPMCI (along with VPCI, VPCI RSI, VPMI, MFI, RSI, VPMI, VPMCI) into a database manager Stata. See the methodology for VPCI, VPMI and VPMCI for their equations.

3. METHODOLOGY

We take all stocks that pass each filter at each decile. We do this for the VPCI, VPMI and VPMCI. We record the annualized returns and standard deviations and coefficient of variation for each decile.

We take all of the stocks that pass the filter and buy them and then rebalance those that drop off assume equal weighting each time. If there is a different number of stocks and we are holding one from the previous rebalance we would add or reduce weight to accommodate the new holding.

a. VPCI

VPCI contrasts price trends to volume-weighted price trends by contrasting the VWMA with corresponding SMAs. VPCI is used to confirm or contradict the price trend. (Volume Analysis).

VPC is found by subtracting the long-term SMA from the same time frame VWMA and is the nexus between price and price that has been proportionally weighted to volume. (Volume Analysis).

VM = volume multiplier (short term volume (13 periods)) / Long term volume (52 periods)). VM overweights VPCI when the volume increases and underweights when it is decreasing (Volume Analysis).

VWMA weighs each period's closing price with that day's volume, divided by the total volume during the period. (Volume Analysis)

VPCI equation: We basically smooth prices out like a moving average but weight on volume and make a couple either tweaks using VWAP.

$$VWMA = \frac{\sum_{i=1}^n Prc_i * Vol_i}{\sum_{i=1}^n Vol_i} \quad (1)$$

$$MA = \frac{\sum_{i=1}^n Prc_i}{n} \quad (2)$$

$$VPR = VWMA(5)/MA(5) \quad (3)$$

$$VolAvg = \frac{\sum_{i=1}^n Vol_i}{n} \quad (4)$$

VPCI is the first indicator we test using deciles.

$$VM = VolAvg(13)/VolAvg(52) \quad (5)$$

$$VPC = VWMA(20)-MA(20) \quad (6)$$

$$VPCI = VPR*VM*VPC \quad (7)$$

A common signal with the VPCI is to take a lower standard deviation bound around the moving average of the VPCI. We use a 20 period moving average and look for values that are extreme for the data series. VBOTs are highly profitable when they occur.

b. VPMI

The VPMI is the difference between internal relative strength and volume weighted relative strength.

To compute VPMI we compute the VPCI RSI first. We then compute the VWMA RSI. The VPCI RSI is the VPCI smoothed by internal relative strength. It is designed to smooth out the VPCI.

VPMI Volume Price Momentum Indicator = $VPMI = 1 + (MFI - RSI)/100$.

$$VPCIUp = VPCI_t - VPCI_{t-1} \text{ if } (VPCI_t - VPCI_{t-1}) > 0 \quad (8)$$

$$VPCIDn = VPCI_{t-1} - VPCI_t \text{ if } (VPCI_{t-1} - VPCI_t) > 0 \quad (9)$$

$$VPCIRS = \frac{\sum_{i=1}^n VPCIUp_i}{\sum_{i=1}^n VPCIDN_i} \quad (10)$$

$$VPCIRSI = (100 - (100/(1 + VPCIRS))) \quad (11)$$

$$VWMAUp = VWMA_t - VWMA_{t-1} \text{ if } (VWMA_t - VWMA_{t-1}) > 0 \quad (12)$$

$$VWMADn = VWMA_{t-1} - VWMA_t \text{ if } (VWMA_{t-1} - VWMA_t) > 0 \quad (13)$$

$$VWMARS = \frac{\sum_{i=1}^n VWMAUp_i}{\sum_{i=1}^n VWMADN_i} \quad (14)$$

$$VWMARSI = (100 - (100/(1 + VWMARS))) \quad (15)$$

The VPMI is then

$$VPMI = VPCIRSI - VWMARSI \quad (16)$$

c. VPMCI

VPMCI is designed to confirm or contradict volume price momentum. It contrasts VPCI momentum to RSI momentum. It compares VPCI RSI with traditional RSI and uses VM and VPMI to confirm entry. VPMI here contrasts the MFI with RSI by subtracting traditional RSI from MFI. MFI is similar to RSI. When MFI goes up, it gains upward price and volume momentum. (Volume Analysis).

VPMCI = Volume Price Momentum Confirmation Indicator $VPMCI = (1 + (MFI - RSI)/100) * 1 + (VPCI - RSI)/100 * VM$

$$TypicalPrice = \frac{High + Low + Close}{3} \quad (17)$$

$$RawMF = \begin{cases} TypicalPrice_t * Vol_t & (TypicalPrice_t - TypicalPrice_{t-1}) > 0 \\ -TypicalPrice_t * Vol_t & (TypicalPrice_t - TypicalPrice_{t-1}) < 0 \end{cases} \quad (18)$$

$$MFR = \frac{\sum_{i=1}^{14} RawMF > 0}{-\sum_{i=1}^{14} RawMF < 0} \quad (19)$$

$$MFI = (100 - 100/(1 + MFR)) \quad (20)$$

$$Up = Close_t - Close_{t-1} \text{ if } (Close_t - Close_{t-1}) > 0 \quad (21)$$

$$Dn = Close_{t-1} - Close_t \text{ if } (Close_{t-1} - Close_t) > 0 \quad (22)$$

$$RS = \frac{\sum_{i=1}^n UP_i}{\sum_{i=1}^n DN_i} \quad (23)$$

$$RSI = (100 - (100/(1 + RS))) \quad (24)$$

$$VPMI = RSI - MFI \quad (25)$$

d. VPMCI

$$VPMCI = VPMI * ((1 + (VPCIRSI - RSI))/100) * VM \quad (26)$$

e. Capital Weighted Volume

Calculating Capital Weighted Dollar Volume in CRSP using daily data. 2000-2021

1. Calculate each components Capital Weighting

a. Market Cap = Price * Shares Outstanding/Factor to Adjust Shares

2. Determine each components Index Weighting

a. Capital Weighting = Stocks Market Cap / Total Market Cap of all Stocks in Sample

3. Calculate each components Capital Weighted Volume

a. Individual Capital Weighted Volume (CWV) = Volume * Capital Weighting

4. Parse Upside VS Downside Volume Tallies by Component

a. Up CWV = Weighted Volume if Individual Stock Price Change >0

b. Down CWV = Weighted Volume if Individual Stock Price Change <0

5. Add Upside + Downside CW Volume = Total Capital Weighted Volume

a. Total Up Capital Weighted Volume = Total Each Stocks Up Capital Weighted Volume

b. Total Down Capital Weighted Volume = Total Each Stocks Down Capital Weighted Volume

6. Subtract Upside – Downside CW Volume & accumulate difference overtime

i. CW Volume Accumulation

7. We take the total up volume - total down volume and get excess up capital weighted volume for accumulating.

8. Then we take the current capital weighted excess up volume and add it to the next value (up or down) and create a running total.

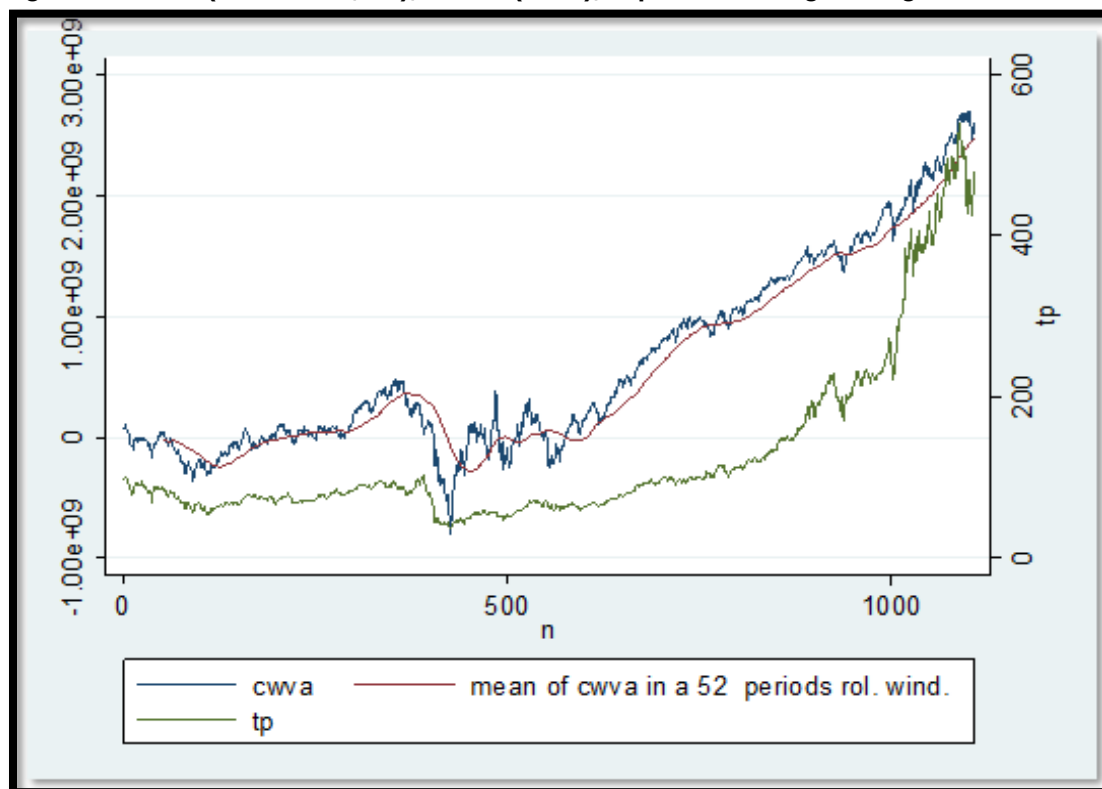
Now we take everything down to an index. We create a crossover strategy using the capital weighted volume. We buy when the accumulated capital weighted volume is above it's 52 day exponential moving average.

We show a graph of the capital weighted volume and its 52 day moving average in the graph below. This shows a snippet of what the indicator looks like with a buy and sell signal above the 52 day average and below the 50 day average.

To adjust the above for Capital Weighted Dollar Volume we **Parse Upside VS Downside Volume Tallies by Component. This involves two changes to steps a and b by taking the product of up stock price changes and volume, and down price changes and volume.**

- a. Up CWV = Weighted Volume * Individual Stock Price Change >0
- b. Down CWV = Weighted Volume * Individual Stock Price Change <0

Figure 2. SP500 (Total Price, TP), CWVA (CWV), 52period Moving Average



The figure above shows us the capital weighted volume accumulated in blue (CWVA), the simple moving average over 52 weeks and the price index for the S&P 500. We report the results for testing the VPCI and Capital Weighted Volume in the results section following.

4. RESULTS

Table 1 Raw Buy and Hold

	Raw
Mean	14.05%
Std Dev	68.46%
CV	20.52%

Table 1 a. VPCI

Decile	10	9	8	7	6	5	4	3	2	1
Mean	5.34%	13.39%	17.17%	16.97%	18.94%	20.56%	22.45%	19.31%	16.40%	13.04%
Std Dev	77.89%	66.80%	65.59%	63.45%	67.82%	63.81%	65.37%	58.09%	60.01%	81.93%
CV	6.86%	20.05%	26.17%	26.75%	27.93%	32.22%	34.35%	33.24%	27.33%	15.91%

Table 1 b. VPMI

Decile	10	9	8	7	6	5	4	3	2	1
Mean	14.41%	14.04%	14.22%	15.14%	14.75%	14.97%	14.81%	15.35%	15.34%	15.00%
Std Dev	71.72%	62.13%	64.73%	62.61%	61.92%	63.89%	62.61%	62.97%	64.93%	63.43%
CV	0.20	0.23	0.22	0.24	0.24	0.23	0.24	0.24	0.24	0.24

Table 1 c. VPMCI

Decile	10	9	8	7	6	5	4	3	2	1
Mean	19.21%	18.79%	18.92%	17.72%	15.71%	14.09%	15.08%	12.30%	10.78%	8.92%
Std Dev	64.78%	59.52%	61.13%	64.63%	67.65%	61.15%	61.66%	61.28%	62.16%	69.74%
CV	0.30	0.32	0.31	0.27	0.23	0.23	0.24	0.20	0.17	0.13

Table 2. Combined Results

	Raw	VPCI Decile 3-5 VPCI	VPMI Decile 6 VPMI	VPMCI Decile 6 VPMCI
Mean	13.98%	21.27%	26.22%	36.86%
Sigma	66.33%	60.61%	63.80%	56.21%
CV	0.21	0.35	0.41	0.66

The first column is the raw return over U.S. listed stocks. The second is the return for the VPCI deciles 3, 4, and 5. It is higher by about 7% and lower risk by about 6%. We then locate only VPCI decile 3-5 stocks within VPMI decile 6. This is done for the previous month to by the stocks in the current month. This improves the return another 5% but adds 3% to the risk. We then add the last condition that the two previous conditions are met within VPMCI decile 6. Our goal is to improve the return while reducing risk. We do this by adding more than 10% return and lower risk by 7.5% from the last sort or 4.5% from the original. This is a model that can be used each month to find stocks that pass and invest for the next month.

The advantage of combining them in a model is that we are picking up on different strengths of the indicators at different levels. Middle trend as measured by the VPCI levels 3-5 works well with moderate momentum VPMI and VPMCI level 6. We are catching stocks before they break into their trend.

We can see from the tables below that the individual VPCI deciles 1-10 are all slightly higher than the raw returns which are the buy and hold. Except for deciles 1 and 9-10. VPMI is all at are above the raw return but not by more than the VPCI. VPMCI is above the raw return in high deciles by itself.

What we see is that combining them we achieve a result better than each indicator can on their own. Combining VPMI 6 with VPCI 3-5 achieves a better result (annualized return 26.22%, risk 63.80%) than either indicator does on its own. Adding another layer, VPMCI level 6 achieves yet a better result than all three do individually with average annualized return of 36.86% and risk of 56.21%. By adding that las layer we improve the return while reducing the overall risk.

We test the capital weighted volume and capital weighted dollar volume on the S&P 500 index in the next table. We expand it further to individual S&P 500 stock constituents from 2002-2022 and then the Center for Research in Security Prices (CRSP) share code 10 and 11 database.

Table 3. SP500 CWV, CWDV 2001-2022

Risk Overlay	CWV	CWDV	SP500
<i>Return</i>	7.65%	6.67%	7.42%
<i>Risk</i>	11.01%	12.55%	17.78%

As a check we test whether capital weighted volume is similar to capital weighted dollar volume on the SP500 over 2001-2022. We then apply the test to the sector SPDR ETFs using Weekly Data for 2002-2022.

Table 4. Weekly CWV SP 500 – 2002-2022

Buy Hold	CWV – 50 Week MA	SP500
<i>Return</i>	9.28%	10.18%
<i>Risk</i>	15.80%	22.93%
<i>CV</i>	0.58	0.44
<i>t-test</i>	2.7144	2.0468
<i>p-value</i>	0.0034	0.0205

Table 5. Daily CWV Center For Research in Security Prices (CRSP) – 2001-2021

	CWV – 250 Day MA	CRSP Share code 10,11
<i>Return</i>	10.05%	9.44%
<i>Risk</i>	12.55%	18.77%
<i>CV</i>	0.80	0.50
<i>t-test</i>	3.68	2.31
<i>p-value</i>	0.0001	0.0104

a. Kolmogorov Smirnov Analysis**Table 6. SP500**

Kolmogorov-Smirnov Test	D	P-value
Rets:	.9290	0.000
Cumulative:	.0677	0.000
Combined K-S:	.9290	0.000

Table 7. CRSP

Kolmogorov-Smirnov Test	D	P-value
Rets:	.9673	0.000
Cumulative:	.0320	0.000
Combined K-S:	.9673	0.000

This test tells us whether the distribution of returns conditioned on $CWVA > EMA50$ comes from the same distribution of returns as the buy and hold (SP500, CRSP). We reject the hypothesis and confirm the returns come from the parent distribution. (SP500, CRSP).

b. CAPM and Fama French Factor Analysis

This shows us the CAPM and Fama and French 3 factor and 5 factor model for excess stock returns. We take the return of our CWV strategy and regress it on the market risk premium, size, value, investment and profitability factors. We report the excess return measured by alpha remaining when all of the factors are used to explain away the excess return. We start with the single factor (CAPM) model which regresses the excess return on the market risk premium.

We then apply it to the Fama and French 3 Factor model which uses two additional factors of size and profitability. We then extend it further to the 5 factor model which uses two more factors of investment and profitability.

Table 8. SP500

CWVA-rf	alpha	Mktrf	Smb	hml
<i>CAPM</i>	14.33*** (1.57)	85.83*** (19.73)		
<i>FF3F</i>	23.06*** (1.68)	124.93*** (21.87)	(12.29) (1.11)	(85.68) (10.20)

** denotes significance at p-value < 0.05 *** denotes significance at p-value < 0.01

Table 9. CRSP

CWVA-rf	alpha	Mktrf	Smb	hml
<i>CAPM</i>	6.08*** (2.66)	85.83*** (47.54)		
<i>FF3F</i>	5.66*** (2.52)	81.96*** (45.52)	37.23*** (10.19)	29.38** (9.54)

** denotes significance at p-value < 0.05 *** denotes significance at p-value < 0.01

Table 10. CRSP 5 Factor Analysis

CWVA- <i>rf</i>	alpha	Mktrf	Smb	hml	Rmw	Cma
<i>FF5F</i>	6.23*** (2.78)	79.18*** (39.72)	27.62*** (7.30)	19.55*** (5.58)	(37.57) (7.72)	26.60*** (4.14)

** denotes significance at p-value < 0.05 *** denotes significance at p-value < 0.01

We show that the S&P 500 CWV strategy has alpha (excess return) of 14.33% annualized and 23.06% annualized for the CAPM and 3 Factor model respectively. We can see that the returns are statistically significant for the 3 factor model and weakly significant for the CAPM.

For the CRSP database we see that the model holds for both CAPM and 3 factor model, as well as the 5 factor model. The excess return is statistically significant for all three models and very strong levels of significance. The excess returns range from 6.08% for CAPM, 5.66% for 3 factor model and 6.23% for the 5 factor model. All returns are annualized.

We then break the strategy down into sector components in the next section using sector SPDR ETF constituents. We also show a VPCI Vbot strategy on the sector SPDRs and combine the two strategies together in the last section.

5. LIMITATIONS

Limitations for this study include availability of data. We use the center for research in security prices which costs around \$50,000 per year for a subscription through WRDS. We also use the index constituents for the S&P 500 which are difficult to obtain.

6. MANAGERIAL IMPLICATIONS

Investors who wish to pursue this strategy would likely follow the S&P 500 and buy when the accumulated capital weighted dollar volume is above it's exponential moving average, and sell when it is below. For the factor model investing, investors would likely have to set up an automated programming interface (API) through their data provider and programming language (e.g. Python). They would then filter stocks each day based on the indicators used in our study. They would lastly hold a portfolio when stocks are in the desired category of VPCI, VPMI, and VPMCI decile levels.

7. CONCLUSION

We show two strategies. A multi factor model combining the strengths of VPCI, VPMI and VPMCI to catch stocks before the break into high VPCI deciles. We achieve a better return per unit of risk as measured by the coefficient of variation by combining all three indicators in a model. We show that by adding layers to the indicator we improve the returns while reducing risk at a level better than each indicator could on its own. We find that middle trend as measured by the VPCI works well with moderate momentum on both VPMI and VPMCI. This catches stocks before they break out.

We show additional risk overlay using the capital weighted volume, and capital weighted dollar volume indicator on the S&P 500 and Center for Research in Security Prices (CRSP) database. We validate the excess returns using the capital asset pricing model, Fama and French 3 and 5 factor models.

We apply a methodology to de-invest when the capital weighted volume is below it's moving average and de-invest when the capital weighted dollar volume is below it's moving average. We return to 100% exposure when the VPCI V bot occurs.

We also use the capital weighted up volume above it's average as a buy signal, and capital weighted down volume below it's average as a buy signal. We add the 30 week VPCI VBOT time stop to the up average signal.

Between the latter two strategies we see that nearly every sector has a benefit of improved risk/reward measured by coefficient of variation. Several sectors perform better on higher return and lower risk, while others enjoy lower return and lower risk and improved risk per unit of return.

For future work a final risk overlay could be applied from the multi factor model to remove the stocks from the portfolio when they are in the 'undesired' category from the overlay. The benchmark for improvement would be improving return or lowering risk by 2%.

We apply the methodology to a random 10 components of all listed stocks from 2015-2022. We see that using a weekly rebalance we perform similar to the sector SPDR ETFs over the same time frame. When we extend the methodology to earlier periods 1999-2022, 2005-2022 the strategy performs similar to the SP500 and CRSP study in this paper. This would lead us to believe the sector SPDR ETF would work with the index constituents and a full sample.

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